

1. Start a transaction and update marks of student with id = 2 to 80, then commit the changes.

```
START TRANSACTION;
```

```
UPDATE students
```

```
SET marks = 80
```

```
WHERE id = 2;
```

```
COMMIT;
```

2. Start a transaction, delete the student whose id = 3, and rollback the transaction.

```
START TRANSACTION;
```

```
DELETE FROM students
```

```
WHERE id = 3;
```

```
ROLLBACK;
```

3. Insert a new student (4, 'Kiran', 88) and commit the transaction.

```
START TRANSACTION;
```

```
INSERT INTO students (id, name, marks)
```

```
VALUES (4, 'Kiran', 88);
```

```
COMMIT;
```

4. Start a transaction, update marks of id = 1 to 95, create a savepoint, then update marks of id = 2 to 60. Rollback only the second update.

```
START TRANSACTION;
```

```
UPDATE students
```

```
SET marks = 95
```

```
WHERE id = 1;
```

SAVEPOINT sp1;

UPDATE students

SET marks = 60

WHERE id = 2;

ROLLBACK TO SAVEPOINT sp1;

COMMIT;

5. Write a query to demonstrate the use of SAVEPOINT and ROLLBACK TO SAVEPOINT.

START TRANSACTION;

UPDATE students

SET marks = 90

WHERE id = 1;

SAVEPOINT sp1;

UPDATE students

SET marks = 70

WHERE id = 2;

ROLLBACK TO SAVEPOINT sp1;

COMMIT;

6. Start a transaction, insert two records, then rollback the entire transaction.

START TRANSACTION;

INSERT INTO students (id, name, marks)

VALUES (5, 'Rahul', 85);

```
INSERT INTO students (id, name, marks)
VALUES (6, 'Anu', 90);
```

```
ROLLBACK;
```

7. Start a transaction, update multiple rows, and commit only if all updates are successful.

```
START TRANSACTION;
```

```
UPDATE students
SET marks = 85
WHERE id = 1;
```

```
UPDATE students
SET marks = 90
WHERE id = 2;
```

```
UPDATE students
SET marks = 95
WHERE id = 3;
```

```
COMMIT;
```

8. What will happen if you execute a ROLLBACK after COMMIT? Write a query to demonstrate this.

```
START TRANSACTION;
```

```
UPDATE students
SET marks = 100
WHERE id = 1;
```

```
COMMIT;
```

```
ROLLBACK;
```

9. Write a TCL query to undo only part of a transaction using SAVEPOINT.

START TRANSACTION;

UPDATE students
SET marks = 75
WHERE id = 1;

SAVEPOINT sp1;

UPDATE students
SET marks = 65
WHERE id = 2;

ROLLBACK TO SAVEPOINT sp1;

COMMIT;

10. Write SQL queries showing the difference between COMMIT and ROLLBACK.

COMMIT

START TRANSACTION;

UPDATE students
SET marks = 90
WHERE id = 1;

COMMIT;

COMMIT: Permanently saves the changes.

ROLLBACK

START TRANSACTION;

UPDATE students
SET marks = 50
WHERE id = 1;

ROLLBACK;